

P-3: The nucleus of a plutonium-239 atom contains 94 protons. Assume that the nucleus is a sphere with radius 6.64 fm and with the charge of the protons uniformly spread through the sphere. At the surface of the nucleus, what are the (a) magnitude and (b) direction (radially inward) of the electric field produced by the protons.

Nucleus contains 94 protons = Z

Radius of the nucleus = 6.64 fm

Given $\rightarrow$ Charge is uniformly dist. over the sphere
(a) Magnitude of Electric field

$$E = \frac{q}{4\pi\epsilon_0 r^2} = \frac{94 \times 1.6 \times 10^{-19} \times 9 \times 10^9}{(6.64 \times 10^{-15})^2}$$

$$E = 3.1 \times 10^{21} \text{ N/C}$$

(b) Direction of the Electric field to the surface will be outward because of the positive charge.

P-5: A charged particle produces an Electric field with a magnitude of 2.0 N/C at a point that is 50 cm away from the particle. What is the magnitude of the particle's charge?

$$E = 2 \text{ N/C}, \quad r = 50 \text{ cm} = 50 \times 10^{-2} \text{ m}$$

$$|q| = ?$$

$$E = \frac{|q|}{4\pi\epsilon_0 r^2} \Rightarrow 2 = \frac{|q| \times 9 \times 10^9}{(50 \times 10^{-2})^2}$$

$$|q| = 5.6 \times 10^{-11} \text{ C}$$

P-7: In Fig 22-35 The four particles form a square of edge length $a = 5.00 \text{ cm}$ and have charges $q_1 = +10.0 \text{ nC}$, $q_2 = -20.0 \text{ nC}$, $q_3 = +20.0 \text{ nC}$ and $q_4 = -10.0 \text{ nC}$. In unit vector notation, what net Electric field do the particles produce at the square's centre?

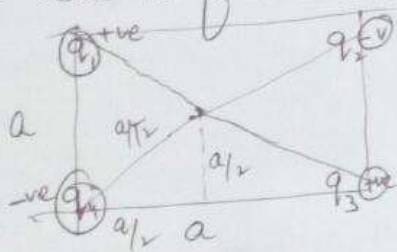
$$\mathbf{E} = E_x \mathbf{i} + E_y \mathbf{j}$$

$$\mathbf{E} = (7.19 \times 10^4) \mathbf{i} + (7.19 \times 10^4) \mathbf{j}$$

Edge length of the square

formed by 4 charged particles $= a = 5 \text{ cm} = 5 \times 10^{-2} \text{ m}$

We have to find Net Electric field at square centre $= E_{\text{net}} = ?$



Each charge is a distance $\frac{\sqrt{2}a}{2} = \frac{a}{\sqrt{2}}$ (half of diagonal) from the square's center.

$$E_x = \frac{1}{4\pi\epsilon_0} \left(\frac{|q_2|}{(a/\sqrt{2})^2} - \frac{|q_4|}{(a/\sqrt{2})^2} \right) \rightarrow \text{x-axis } q_2 \text{ and } q_4$$

$$= \frac{1}{4\pi\epsilon_0} \left(\frac{2q}{a^2/2} - \frac{q}{a^2/2} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{a^2/2} \right)$$

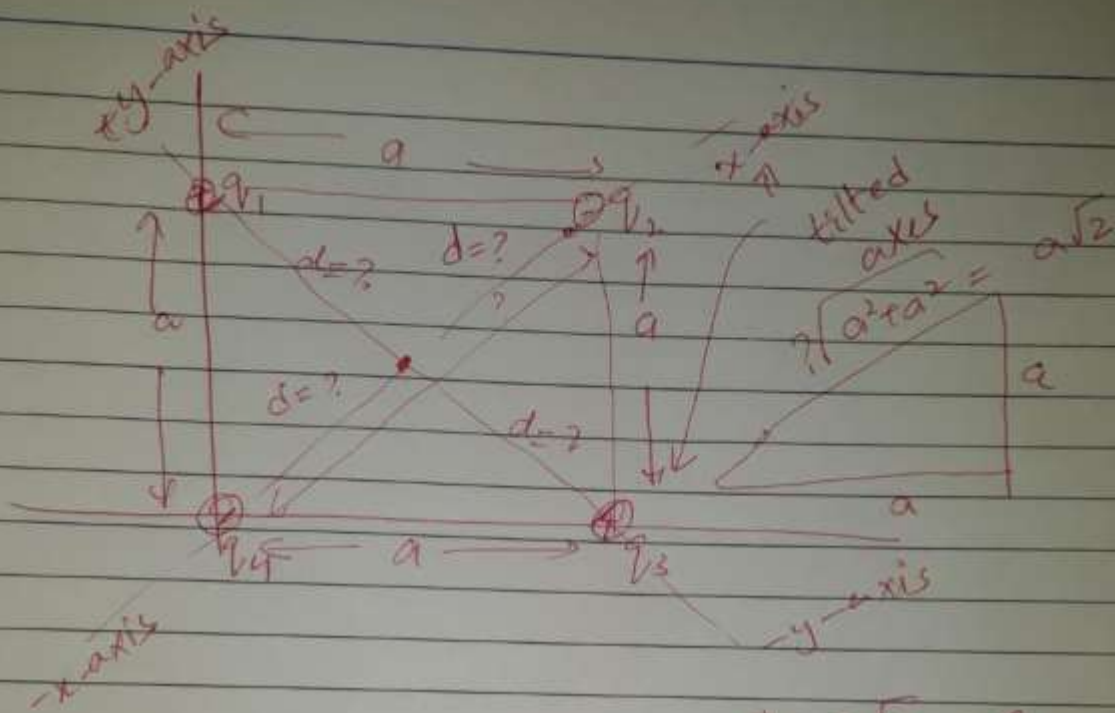
$$= \frac{9 \times 10^9 \times 10 \times 10^{-9}}{(5 \times 10^{-2})^2/2} = 7.19 \times 10^4 \text{ N/C}$$

Similarly: $E_y = 7.19 \times 10^4 \text{ N/C}$. y-axis q_1 & q_3

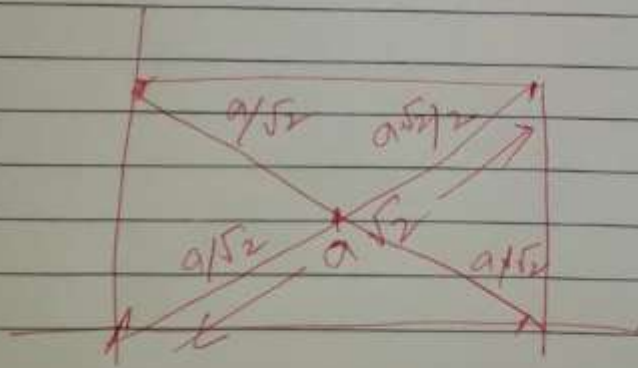
$$\text{So, } E = \sqrt{E_x^2 + E_y^2}$$

$$= \sqrt{(7.19 \times 10^4)^2 + (7.19 \times 10^4)^2}$$

$$E = 1.02 \times 10^5 \text{ N/C}$$



$$d = \frac{a\sqrt{2}}{2} = \frac{a}{\sqrt{2}}$$



TOPIC: Electric field due to a line of charge.

P-22: Density, density, density, (a) A charge -300e is uniformly distributed along circular arc of radius 400cm , which subtends an angle of 40° . What is the linear charge density along the arc? (b) A charge -300e is uniformly distributed over one face of a circular disk of radius 2.00cm . What is the surface charge density over the face? (c) A charge -300e is uniformly distributed over the surface of sphere of radius 2cm . What is the surface charge density over that surface? (d) A charge -300e is uniformly spread through the volume of a sphere of radius 2cm . What is the volume charge density?

(a) $q = -300\text{e}$ $q = -300\text{e}$
circular arc (radius) = 4cm
subtends at an angle = $\theta = 40^\circ$

$\lambda = ?$ linear charge density

$$\lambda = \frac{q}{l}$$

$$s = r\theta$$

$$s = l = 0.0279\text{m}$$

$$q = -300 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-17}\text{C}$$

$$\lambda = \frac{4.8 \times 10^{-17}}{0.0279} = 1.72 \times 10^{-15}\text{C/m}$$

to convert to degrees into radians, multiply with $\pi/180$:

$$\theta = (40 \times 3.14) / (180)$$

$$\theta = 0.691\text{ rad}$$

(b) $q = -300\text{e}$ over the surface of disk

radius of the disk = $r = 2\text{cm}$

$\sigma =$ surface charge density

$$\sigma = q/A$$

$$\text{Area} = \pi r^2 = 3.14 \times (2 \times 10^{-2})^2 = 1.256 \times 10^{-3}\text{m}^2$$

$$\sigma = \frac{-300 \times 1.6 \times 10^{-19}}{1.256 \times 10^{-3}}$$

$$\Rightarrow \sigma = -3.82 \times 10^{-14}\text{C/m}^2$$

(c) $q = -300e$
charge is distributed over surface of a sphere
of radius 2 cm

$$\text{Area of the sphere} = 4\pi r^2$$

$$= 4 \times 3.14 \times (2 \times 10^{-2})^2$$

$$\sigma = \frac{q}{A} = \frac{-300 \times 1.6 \times 10^{-19}}{4\pi r^2}$$

$$\Rightarrow \sigma = -9.5 \times 10^{-15} \text{ C/m}^2$$

(d) $q = -300e$ spread through volume of a sphere
of radius 2 cm

$$\rho = \frac{q}{\text{Vol.}} \quad \rho = \frac{q}{\frac{4}{3}\pi r^3}$$

$$\rho = \frac{-300 \times 1.6 \times 10^{-19}}{\frac{4}{3}\pi r^3}$$

$$= \frac{-4.8 \times 10^{-17}}{\frac{4}{3} \times 3.14 \times (2 \times 10^{-2})^3}$$

$$\rho = -1.43 \times 10^{-12} \text{ C/m}^3$$

P-42: Humid air breaks down (its molecules become ionized) in an Electric field of $3.0 \times 10^6 \text{ N/C}$. In that field what is the Magnitude of the electrostatic force on (a) an electron (and) (b) an ion with a single electron missing?

$$E = 3 \times 10^6 \text{ N/C}$$

$$F = ? \text{ on } e^- \text{ (a)}$$

$$\text{on ion Missing } e^- \text{ (b)}$$

$$(a) F_e = eE = 1.6 \times 10^{-19} \times 3 \times 10^6 = 4.8 \times 10^{-13} \text{ N}$$

$$(b) F_i = Eq = 3 \times 10^6 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-13} \text{ N}$$

P-43: A electron is released from rest in uniform electric field of magnitude $2 \times 10^4 \text{ N/C}$. Calculate the acceleration of the e^- . (ignore gravitation)

$$E = 2 \times 10^4 \text{ N/C}$$

$$\text{acc. of the electron} = ?$$

$$a = F/m = \frac{eE}{m} = \frac{1.6 \times 10^{-19} \times 2 \times 10^4}{9.11 \times 10^{-31}}$$

$$a = 3.51 \times 10^{15} \text{ m/s}^2$$

P-44: An Alpha particle (The nucleus of a helium) has a mass of $6.64 \times 10^{-27} \text{ kg}$ and charge of $+2e$. What are the (a) magnitude and (b) direction of the electric field that will balance the gravitational force on the particle?

$$F_E = F_g$$

$$qE = mg \Rightarrow E = \frac{mg}{q} \Rightarrow E = \frac{mg}{2e}$$

For α -particle $q = +2e$

$$E = 2 \times 10^{-7} \text{ N/C}$$

P-47: Beams of high speed protons can be produced in 'guns' using electric fields to accelerate the protons. (a) What acceleration would a proton experience if the gun's electric field are $2.00 \times 10^4 \text{ N/C}$? (b) What speed would the proton attain if the field accelerated the proton through a distance of 1.00 cm ?

(a) acceleration of proton = ?

Electric field of the gun $= E = 2 \times 10^4 \text{ N/C}$

(b) speed of proton = ?

Through a distance of 1 cm

$$a = \frac{F}{m} = \frac{qE}{m} = \frac{1.6 \times 10^{-19} \times 2 \times 10^4}{1.67 \times 10^{-27}}$$

$$a = 1.9 \times 10^{12} \text{ m/s}^2$$

(b) Suppose proton was at rest initially

$$v^2 = v_0^2 + 2ad$$

$$v_0 = 0 \Rightarrow v = \sqrt{2ad}$$

$$= \sqrt{2 \times 1.9 \times 10^{12} \times 1 \times 10^{-2}}$$

$$v = 1.96 \times 10^5 \text{ m/s}$$